

Accurate Metal–Imidazole Interactions

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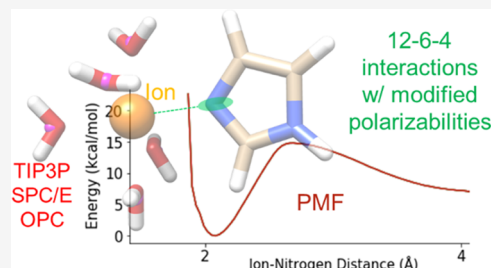


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ABSTRACT: Modeling the interaction between a metal ion and small molecules can provide pivotal information to bridge and close the gap between two types of simulations: metal ions in water and metal ions in metalloproteins. As previously established, the 12-6-4 Lennard-Jones (LJ)-type nonbonded model, because of its ability to account for the induced dipole effect, has been highly successful in simulating metal ion systems. Using the potential of mean force (PMF) method, the polarizability of the metal-chelating nitrogen from two types of imidazole molecules, delta nitrogen protonated (HID) and epsilon nitrogen protonated (HIE), has been parametrized against experiment for 11 metals (Ag(I), Ca(II), Cd(II), Co(II), Cu(I), Cu(II), Fe(II), Mg(II), Mn(II), Ni(II), and Zn(II)) in conjunction with three commonly used water models (TIP3P, SPC/E, and OPC). We show that the standard 12-6 and unmodified 12-6-4 models are not able to accurately model these interactions and, indeed, predict that the complex should be unstable. The resultant parameters further establish the flexibility and the reliability of the 12-6-4 LJ-type nonbonded model, which can correctly describe three-component interactions between a metal, ligand, and solvent by simply tuning the polarizability of the chelating atom. Also, the transferability of this model was tested, showing the capability of describing metal–ligand interactions in various environments.



1. INTRODUCTION

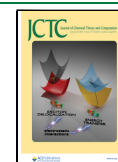
Metal ions are essential elements for the maintenance of plants, animals, and humans.^{1–3} The absence of specific metal ions can result in growth disorders, carcinogenesis, severe malnutrition, or death. Over 25% of proteins contain metal ions, which function in either a structural or catalytic role.^{4–6} Computational chemistry has become an effective tool to investigate metal ion-coordinating systems present in various biological systems (such as proteins, nucleic acids, carbohydrates, and lipids).^{7–11} When compared to quantum mechanics-based models, force field models have significant advantages in computational cost. However, reproducing the structural features (e.g., ion–oxygen distance (IOD)) and thermodynamics (e.g., hydration free energy (HFE)) of metal ions in water or protein systems is a challenging task.¹² Depending on the system and the research question, various force field models are available; for instance, the bonded,¹³ nonbonded,¹⁴ Drude oscillator,^{15,16} cationic dummy atom (CDA),¹⁷ CDA_{pol},¹⁸ and ReaxFF models.¹⁹ The 12-6 Lennard-Jones (LJ) nonbonded model is used extensively because it has a simple form, is easy to parametrize, and has excellent transferability.^{17,20,21} Nevertheless, the 12-6 LJ nonbonded model fails to reproduce the experimental ion–oxygen distance (IOD) and hydration free energy (HFE) of the first solvation shell at the same time for many ions due to an underestimation of the ion–water interaction energy.¹² This can be compensated for by scaling the LJ parameters in unphysical ways,^{22,23} which in our work, we have studiously avoided by working with physically reasonable parameter ranges.²⁴ The

deficiency of the 12-6 model is primarily because it does not include ion-induced dipole interactions, which should not be neglected in highly polarized systems.²⁵

In the bonded model, the metal ion is covalently bound to the coordinated residues, in which the bond angle, dihedral, van der Waals, and electrostatic interactions are defined by classical terms. Several schemes have been devised to parametrize these terms. Despite the success of the bonded model in replicating the experimentally determined structures, it cannot simulate the coordination number or ligand changes, which is crucial to model catalytic metal centers and metal transport. The nonbonded model is another popular model for metal ions, where the metal ion is represented by a soft sphere that interacts with the surroundings through vdW and electrostatic interactions. The vdW interactions are defined by the 12-6 LJ²⁶ or Born–Mayer potential.²⁷ In the 12-6 LJ model, the C_4 term was introduced by Li, Merz, and co-workers to account for the ion-induced dipole interaction. The ion-induced dipole interaction is proportional to r^{-4} , where r is the distance between the two particles.²⁵ Therefore, the new model is called the “12-6-4 nonbonded model”, which can

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successfully reproduce the experimental HFE and IOD simultaneously for various metal ions.²⁵

Since the 12-6-4 model is easy to apply, is computationally efficient, and accurately describes the interactions between the metal ion and its coordinating ligands, it serves as an excellent model for simulating metal ions in molecular dynamics (MD) simulations. In earlier research,²⁸ we only studied Co²⁺ and Ni²⁺ interactions with delta nitrogen protonated (HID)-charged histidine in TIP3P water. In the present study, we have been more systematic, including 11 ions (Ag(I), Ca(II), Cd(II), Co(II), Cu(I), Cu(II), Fe(II), Mg(II), Mn(II), Ni(II), and Zn(II)) with both HID- and epsilon nitrogen protonated (HIE)-charged histidines and in three water models: TIP3P, SPC/E, and OPC.²⁹ By tuning the C₄ term in the 12-6-4 model, we were able to successfully match related experimental interaction energies for each of the “ion–imidazole and charge–water model” combinations. This work will benefit metalloprotein research significantly since these 11 ions are reported to have up to 2992 occurrences of interactions with histidine(s) in the PDB databank (Table S1).

The present work provides a validated set of His-metal ion interactions that can be used to study, with a higher degree of confidence than before, metal ions coordinated by His in proteins. This will facilitate metalloprotein design, understanding the mechanism of transition-metal ion transport, and metal ion selectivity in protein systems. The provided models are compatible with the AMBER force fields (ff14SB, ff19SB) and several water models allowing a range of applications.

2. COMPUTATIONAL METHODS

2.1. Optimization of the 12-6-4 Potential. In this work we have used a 12-6-4 nonbonded model along with the AMBER force field

$$\begin{aligned}
 U_{ij}(r_{ij}) &= \frac{C_{12}^{ij}}{r_{ij}^{12}} - \frac{C_6^{ij}}{r_{ij}^6} - \frac{C_4^{ij}}{r_{ij}^4} + \frac{e^2 Q_i Q_j}{r_{ij}} \\
 &= \varepsilon_{ij} \left[\left(\frac{R_{\min,ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{R_{\min,ij}}{r_{ij}} \right)^6 \right] - \frac{C_4^{ij}}{r_{ij}^4} + \frac{e^2 Q_i Q_j}{r_{ij}} \\
 &= 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] - \frac{C_4^{ij}}{r_{ij}^4} + \frac{e^2 Q_i Q_j}{r_{ij}}
 \end{aligned} \quad (1)$$

where e depicts the proton charge, while Q_i and Q_j represent the partial charge of atoms i and j , respectively. The Coulomb pair potential is utilized to represent the electrostatic interaction between atoms i and j , while the classic 12-6 LJ potential plus an extra r^4 term represents the van der Waals interactions. The C_4 terms between water and ions are taken from our previous studies.^{14,25,30} The C_4 terms between the histidine and metal ions are optimized based on the following equation

$$U_{q-\alpha_0} = -\frac{1}{2}\alpha_0 \left(\frac{q}{4\pi\epsilon_0\epsilon_r r^2} \right)^2 \cos\theta = -\frac{C_4^{ij}}{r^4} \quad (2)$$

where α_0 is an atom-type-dependent polarizability. The metal binds to water molecules and imidazole (which mimics the side chain of His). Potential of mean force (PMF) calculations are used to optimize the pairwise parameters to reproduce the

experimental free energies of each metal bound to imidazole. The HID- and HIE-charges used on imidazole molecules are described in Figure 1, which are the same as what ff14sb and

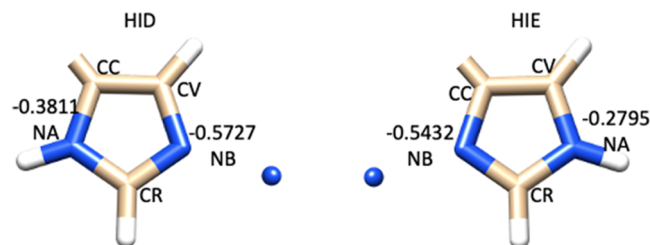


Figure 1. Illustration and comparison of the charge distribution for both HID and HIE imidazole molecules. Heavy-atom-type names in AMBER are also marked.

ff19sb uses in AMBER 20.³¹ The major differences between HID- and HIE-charged histidines are the nitrogen charge values and the protonation locations. To remove uncertain interactions from amine and carboxyl groups from histidine, the connecting carbon (termed CC in Figure 1) is not connected to the α -carbon of histidine but rather a hydrogen in the present work. This has been shown to be a reasonable approximation²⁸ since in real protein simulation, the backbone of histidine is usually rigidly connected to the protein and will not interfere with the metal–ligand interaction like free amino acid does.

2.2. Parametrization Processes. A PMF simulation protocol was prepared according to the MD procedure described in Section 2.3. The parametrization was conducted in an iterative manner. For each iteration, a PMF with a short umbrella sampling (termed us1) of 1 ns for each window was conducted first to calculate approximate interaction energies. When the PMF-calculated interaction energy was ± 0.25 kcal/mol within the experimental binding free energy, a final round of PMFs was conducted with longer umbrella sampling (termed us2) of 3 ns for each window to get a precise estimate of the calculated interaction free energy. If us1 or us2 did not give a binding free energy within the range of ± 0.25 kcal/mol of the experimental value (generally rare since us2 is based on a successful us1), a new round of us1 would be initiated with a newly assigned polarizability value to continue the iteration, until an accurate binding free energy was obtained.

2.3. Molecular Dynamics Simulations. The CUDA version of PMEMD from the AMBER 20³¹ package was utilized to perform all of the molecular dynamics (MD) simulations. The minimization of the system was performed in three steps: (a) minimization of water molecules with the imidazole group and ions restrained, (b) minimization of side-chain hydrogens (which are imidazole hydrogens) on the nonwater molecules, and (c) minimization of the whole system. Steps (a) and (b) consists of 10 000 cycles of minimization using the steepest descent method, followed by 10 000 cycles using the conjugate gradient method. Meanwhile step (c) yields three “steepest descent then conjugate gradient” cycles as mentioned in steps (a) and (b). In the next step, the system was heated to 300 K gradually during a 1 ns NVT simulation. For system equilibration, a 550 000-step run was done at 300 K employing the NPT ensemble. Finally, a production run for 500 000 steps at 300 K under constant NPT conditions was performed. The Langevin thermostat with

a collision frequency of 1 ps^{-1} was applied to control the temperature, and the Berendsen barostat³² with a pressure relaxation time of 5 ps was employed for the pressure control. The time step was 2 fs, and the nonbonded cutoff was 10 Å. The SHAKE³³ algorithm was used to constrain bonds involving hydrogen atoms, and the time step was set to 2 fs. Cluster analysis was utilized to obtain the most representative structure from the MD simulations. The UCSF Chimera³⁴ and VMD programs were used for the visualization and preparation of the figures shown in this study.

3. RESULTS AND DISCUSSION

3.1. Overview of Target Interaction Energies, Ion Properties, and Parametrization Results. The 12-6-4 parameter set was used to reproduce the experimental metal–imidazole binding free energies compared against experimental results. The target binding free energies computed from experimental log K values are shown in Table 1.³⁵ Amongst all

Table 1. Experimental Interaction Energy of Each Ion

ions	electronic configuration	effective ionic radius ⁴² (Å)	energy ³⁵ (kcal/mol)
Ag(I)	[Kr]4d10	1.15	−3.98
Ca(II)	[Ar]	1.00	0.16
Cd(II)	[Kr]4d10	0.95	−3.63
Co(II)	[Ar]3d7	0.75	−3.36
Cu(I)	[Ar]3d10	1.40	−7.89
Cu(II)	[Ar]3d9	0.73	−5.70
Fe(II)	[Ar]3d6	0.78	−2.46
Mg(II)	[Ne]	0.72	−0.10
Mn(II)	[Ar]3d5	0.83	−1.64
Ni(II)	[Ar]3d7	0.69	−4.12
Zn(II)	[Ar]3d10	0.74	−3.48

of the experimental values, Ag(I), Cu(II), Fe(II), Mn(II), Ni(II), and Zn(II) were obtained from thermometric titration;^{36–38} Ca(II) and Mg(II) were obtained from indicator titrations;³⁹ Co(II) and Cd(II) were obtained from potentiometric titration with perchloric acid;⁴⁰ while Cu(I) was obtained from electron absorption spectrum during ligand exchange.⁴¹ The experimental free energies of interactions of each ion show a high dependence on both the ionic radius and the electronic configuration. Overall, the larger effective ionic

radius ions have the lower binding free energy with imidazole molecules with a couple of exceptions in the transition-metal series. For these exceptions, when the d-orbitals of the ions are full or half full, they tend to have higher, meaning less negative binding free energies with imidazole molecules, indicating a connection between the d-orbital symmetry and imidazole interaction energies.³⁵

After investigating the correlation between ion properties and their interaction free energies with imidazole molecules, the parametrization process was conducted following the protocol summarized in the Computational Methods section. The results agree well with previous studies⁴³ and what Table S2 shows (Table S2 displays all of the default values), where the current 12-6-4 LJ model globally underestimates the interaction. Therefore, higher polarizability values for nitrogen are needed. However, according to the parametrized polarizability data displayed in Table 2, it is interesting to observe that a higher experimental interaction energy does not always correspond to higher polarizability on nitrogen. This inconsistency between binding free energies and polarizabilities indicates that the interaction between a metal ion and an imidazole is a result of both the dipole-inducing ability of the metal ion and the capability of the imidazole nitrogen to respond to that induced dipole. The first dipole-inducing ability is solely a feature of the metal ion itself, while the second response depends on both the imidazole nitrogen and the aqueous environment. Hence, we parametrized the polarizability of nitrogen based on the dependences of both water models and imidazole charges, as displayed in Table 2. Here, three common water models (TIP3P, SPC/E, and OPC) and two common AMBER imidazole charges (HID and HIE) are applied to form six combinations for parametrization.

3.2. Parameter Comparison between Different Water Models and Imidazole Charges. Further analysis of the parametrized polarizabilities from Table 2 suggests that the imidazole ring with the HIE charge (Figure 1; epsilon nitrogen protonated) of −0.5432 needs a higher polarizability value than the imidazole ring with the HID charge (Figure 1; delta nitrogen protonated) of −0.5727, meaning a stronger C₄ interaction is required for the HIE-charged imidazole than the HID-charged imidazole. Therefore, to reproduce the experimental value with less electron density, the HIE-charged imidazole needs to have a higher polarizability value on the chelating nitrogen to match the same experimental value.

Table 2. Parametrized Polarizability (Å³) of Nitrogen and the Final C4 Value (kcal/(mol Å⁴)) between the Metal Ion and Nitrogen to Reproduce the Binding Free Energy Using PMF Studies

ions	TIP3P				SPC/E				OPC			
	HID		HIE		HID		HIE		HID		HIE	
	Pol	C4	Pol	C4	Pol	C4	Pol	C4	Pol	C4	Pol	C4
Ag(I)	4.65	267	5.05	290	4.52	288	4.55	290	4.23	243	4.63	266
Ca(II)	3.75	226	4.40	265	3.71	229	3.77	232	3.68	219	3.69	220
Cd(II)	3.75	483	3.95	509	3.50	482	3.72	512	3.20	485	3.25	499
Co(II)	3.10	451	3.18	462	3.00	434	3.15	456	2.90	410	3.03	428
Cu(I)	29.3	142	30.0	145	22.3	139	24.6	153	13.3	147	13.5	149
Cu(II)	2.93	590	3.05	615	2.89	608	2.99	629	2.75	554	2.79	562
Fe(II)	3.25	367	3.55	401	3.41	366	3.75	403	3.11	332	3.35	357
Mg(II)	3.25	299	3.48	320	3.39	286	3.47	293	2.90	255	3.05	268
Mn(II)	3.41	345	3.68	372	3.46	371	3.50	376	3.00	364	3.03	367
Ni(II)	3.00	442	3.11	459	3.10	440	3.30	468	2.75	403	2.95	433
Zn(II)	2.94	472	3.05	490	2.95	472	3.09	494	2.71	422	2.87	447

Table 3. Properties of All of the Three Water Models Parametrized in This Research

water model	OH bond length	OD bond length	HOH bond angle	oxygen/dummy charge	hydrogen charge	VDW σ	VDW ϵ
TIP3P ⁴⁴	0.9572	N/A	104.72	−0.8340	0.4170	3.1506	0.6364
SPC/E	1.0000	N/A	109.47	−0.8476	0.4238	3.166	0.650
OPC ⁴⁶	0.8724	0.1594	103.6	−1.3582	0.6791	3.1666	0.8903

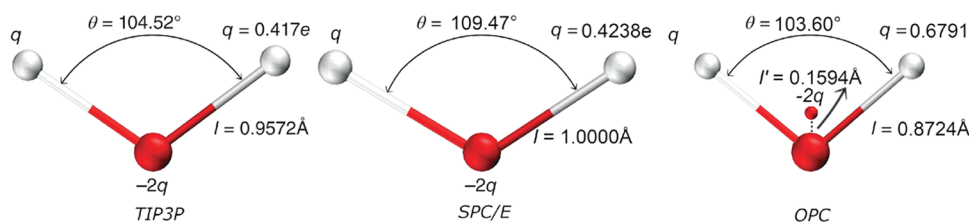
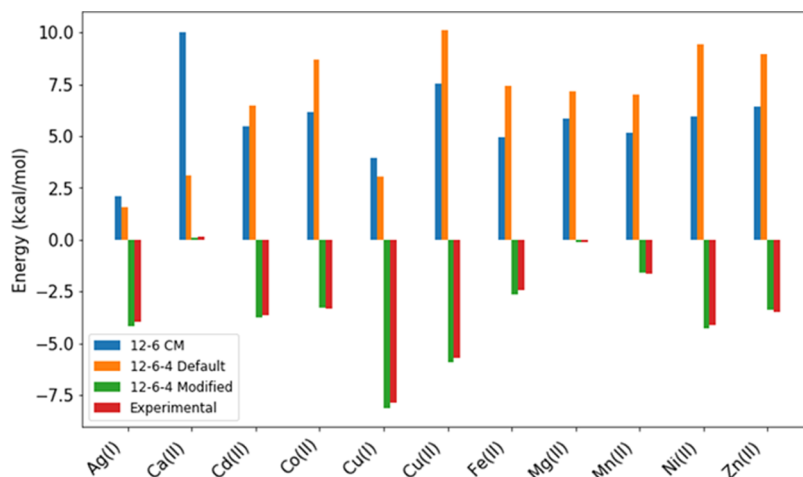
Figure 2. Images of the structure of TIP3P,⁴⁴ SPC/E,⁴⁵ and OPC⁴⁶ water models.

Figure 3. Comparison between experimental binding free energies and calculated binding free energies generated by the 12-6 CM parameter set, default 12-6-4 set, and modified 12-6-4 set for 11 metal ions. Related raw data is presented in Table S3.

Amongst the different water models, the polarizability shows a trend where the polarizabilities of the TIP3P and SPC/E water models are similar, while the polarizability of the OPC water model is lower than those of the TIP3P and SPC/E water models (“TIP3P $\approx$ SPC/E $>$ OPC”). This is an interesting ranking because it shows a strong correlation between the intrinsic properties of three/four-point water models and the parametrized polarizability of the imidazole nitrogen. As shown in Table 3, both the TIP3P and SPC/E models are three-point water models, and they also have larger H–O–H angles and longer O–H bond lengths in comparison to the four-point OPC water model. The unique geometry of OPC allows the metal ion to experience less steric hindrance when one water molecule in the first hydration shell is replaced by a larger imidazole molecule. Therefore, the binding of imidazole in a water shell formed by OPC is thermodynamically more favored than in water shells formed by TIP3P or SPC/E, meaning the interaction between the metal ion and imidazole are, by nature, stronger in OPC water. In short, the OPC water model does not require a higher polarizability value on the imidazole nitrogen to reproduce the same experimental interaction energy relative to TIP3P and SPC/E (Figure 2).

3.3. Parameter Comparison between Different Ions.

Among all of the 11 ions parametrized in this work, their C_4 -dependent polarizabilities showed a strong dependence on both the ion charges and electron configurations. First, it can

be concluded that if the ions are monovalent, like Ag(I) and Cu(I), the chelating nitrogen needs higher polarizabilities, meaning higher C_4 values to reproduce the experimental values. This indicates a larger underestimation of the dipole-inducing ability of monovalent ions on ligands compared to the smaller underestimation on divalent ions. The phenomenon can be explained theoretically using crystal field theory. Being different from water, the imidazole nitrogen is considered a strong-field ligand that will enlarge the energy gap between the e_g and t_{2g} orbitals and push more electrons to the lower-energy t_{2g} -orbital. This will break the d-orbital symmetry and the electrostatic screening effect, thereby enhancing the dipole-inducing abilities of the metal ions. However, this enhancement is not obvious on the only two monovalent ions Ag(I) and Cu(I) because both have filled d-orbitals, *i.e.*, both the e_g and t_{2g} orbital groups are fully occupied. As a result, Ag(I) and Cu(I) still have strong d-orbital screening (meaning the metal ion charge cannot “reach” and “induce” the nitrogen atom effectively), so the parametrized polarizability of nitrogen needs to be larger; in this way, the nitrogen atoms can be more “responsive” to the weak inducing power from the metal ion. This hypothesis also matches the fact that both Ag(I) and Cu(I) are soft acids, which can have a strong induced dipole effect.⁴⁷ Further comparison on the equilibrium distances between Ag(I)–N (2.35 Å) and Cu(I)–N (1.86 Å) reveals

that they tend to have closer metal–nitrogen distances compared to other ions.

For the other nine divalent ions, the rule mentioned above, *i.e.*, “a symmetric d-orbital group leads to a higher nitrogen polarizability,” is not as obvious as for the monovalent ions, likely due to the high charge and small radii of the divalent ions, especially on Zn(II). However, for large divalent ions like Fe(II) and Cd(II), they still yield slightly higher polarizabilities on the nitrogen compared to other ions because for Fe(II), its t_{2g} orbital group is symmetric with six electrons filling three orbitals, while Cd(II) has fully filled d-orbitals.

Lastly, by comparing ions of the same family, *e.g.*, Mg(II) and Ca(II), Zn(II) and Cd(II), Cu(I) and Ag(I), it is interesting to draw the conclusion that for monovalent ions, a larger radius will lead to a lower nitrogen polarizability, while for divalent ions, a larger radius will lead to a higher nitrogen polarizability. This is because the induced dipole decays faster for divalent ions than for monovalent ions.²⁵ When the ion is divalent, a larger ion radius will significantly reduce the distortion of the nitrogen atom on the imidazole, so it needs a higher polarizability to counter this decay. However, when the ion is monovalent, a larger radius will not significantly reduce the dipole-inducing ability, while a smaller radius would in turn overestimate the electron–electron repulsion between the metal ion and nitrogen. To balance the overestimation of the repulsion, Cu(I) needs to have its chelating nitrogen be very polarizable to reproduce the experimental interaction energy.

3.4. Energy Comparison between 12-6 CM, 12-6-4 Default, and 12-6-4 Modified Using HID in OPC. As Figure 3 indicates, the modified parameters generated energy values (green bars) that agree well with the experimental values (red bars). However, we have discovered that the default 12-6-4 LJ parameter set (orange bars) underperformed compared to the 12-6 compromise (CM) set (blue bars) except for Ca(II) and Cu(I). This indicates an underestimation of the aqueous C_4 parameter while being applied in the protein systems; so according to eq 1, the C_4 contribution is less negative, making the total binding free energy more positive. To explain this phenomenon, eq 2 can be used in support because in the ion–water–imidazole system, the dielectric constant ϵ_r (≈ 27)⁴⁸ is smaller than that in the aqueous system (≈ 80) due to the existence of the point charge of the metal ion and the electron cloud of nitrogen. Therefore, if the polarizability is kept small ($1.09 \text{ \AA}^3$) at the numerator but the denominator has been reduced, the whole C_4 term will have a significant underestimation. Then according to eq 1, the contribution of the C_4 term will be less negative, leading to a more positive final interaction energy, as the orange bars shown in Figure 3. This can be fixed by increasing the polarizability of the chelating nitrogen *via* parametrization to fit the environment of the metal binding site. Previous work⁴⁹ suggested that a decorrelation between two 12-6-LJ parameters, $R_{\min}/2$ and ϵ , is able to reproduce the hydration free energy of metal ions in aqueous systems, but this 12-6 model was not adopted herein because of our desire to keep the parameters physically meaningful.

4. CONCLUSIONS

In the present work, we have parametrized the 12-6-4 Lennard-Jones (LJ) nonbonded model for 11 metal ions for delta nitrogen protonated (HID)- and epsilon nitrogen protonated (HIE)-charged histidines for three water models (*i.e.*, TIP3P, SPC/E, and OPC). The final parametrized polarizabilities can

successfully reproduce experimental interaction free energy. Moreover, the result shows a strong correlation between the polarizability of the imidazole nitrogen, geometry of the water model, and the electronic configuration of the metal. This means that, by merely tuning how polarizable the chelating nitrogen is, the model can rationally reproduce the three-component interaction between imidazole, water, and metal. Since this work has successfully extended the reliability of the 12-6-4 LJ nonbonded model from an aqueous system to a metal–ligand system, we are in a position to tackle related problems in biology and chemistry. Overall, the developed metal–imidazole parameter sets will provide the basis for the research community to conduct accurate MD simulations of metalloproteins at a higher level of accuracy than heretofore possible.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.2c01081>.

Count of occurrence where each ion has at least one histidine in its binding sphere of any protein in PDB (Table S1); default energy (kcal/mol) using the default polarizability ($1.09 \text{ \AA}^3$) of nitrogen for each ion to reproduce binding free energy using PMF studies (Table S2); raw data to plot Figure 3 (Table S3) (PDF)

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Notes

The authors declare no competing financial interest.

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